2. Common Elements Between Two Arrays

Aim: To find the common elements between two arrays and analyze their time complexity.

C

```
#include <stdio.h>
int main() {
   int a[100], b[100];
   int n, m, i, j, found;
   printf("Enter size of first array: ");
   scanf("%d", &n);

   printf("Enter first array: ");
   for (i = 0; i < n; i++) {
      scanf("%d", &a[i]);
   }

   printf("Enter size of second array: ");
   scanf("%d", &m);
   printf("Enter second array: ");
   for (i = 0; i < m; i++) {
      scanf("%d", &b[i]);
   }
   printf("Common elements: ");
   for (i = 0; i < n; i++) {
      found = 0;
      for (j = 0; j < m; j++) {
         if (a[i] == b[j]) {
            found = 1;
```

```
    break;
        }
      }
      if (found) {
        printf("%d ", a[i]);
      }
    }
    return 0;
}
```

Input: enter size of first array:5

1 2 3 5 8

4 6 2 8 5

Output:

```
Enter size of first array: 5
Enter first array: 1 2 3 5 8
Enter size of second array: 5
Enter second array: 4 6 2 8 5
Common elements: 2 5 8
```

Result: Thus, the C program successfully finds and displays the common elements between two arrays.